

Last Time:

- V v.s.

(a) $\beta = (v_1, \dots, v_n) \in V^n$ is a basis of V if β l.i. and $\text{span}(\beta) = V$.

(b) $\beta \subseteq S$ of V is a l.i. subset that also spans V .

(two defs agree for $(v_1, \dots, v_n)$ & $\{v_1, \dots, v_n\}$)
ordered unordered

- Std basis of $\mathbb{R}^n$ is $e_1, \dots, e_n$, where

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$$

Today:

Continue §1.4 Basis and Dimension (Properties of basis)

Rmk: By def, if V v.s., $v_1, \dots, v_n \in V$ l.i., then $v_1, \dots, v_n$ is a basis of $\text{span}(v_1, \dots, v_n)$.

Ex 1.47: $\underbrace{\{x^n : n \in \mathbb{Z}_{\geq 0}\}}_{\substack{\infty\text{-set} \\ \text{infinite-set}}} \text{ is a basis of } \mathbb{R}[x].$
 $\{1, x, x^2, x^3, \dots\}$

Pf: Unit 1.4 notes (only for Math-Major).

□

Properties of bases:

Prop 1.48: V v.s. Then $v_1, \dots, v_n \in V$ is a basis iff
 $\forall w \in V$, $\exists!$ l.c. $w = a_1 v_1 + \dots + a_n v_n$.
there exists unique

Pf: $(\Rightarrow)$: Assume that $v_1, \dots, v_n$ is a basis for V .

Then $w \in V = \text{span}(v_1, \dots, v_n)$, so $\exists (a_1, \dots, a_n) \in \mathbb{R}^n$ s.t.

$$w = \sum_{\bar{i}=1}^n a_{\bar{i}} v_{\bar{i}}. \text{ If } \exists (b_1, \dots, b_n) \in \mathbb{R}^n \text{ s.t. } w = \sum_{\bar{i}=1}^n b_{\bar{i}} v_{\bar{i}},$$

$$\text{then } \sum_{\bar{i}=1}^n a_{\bar{i}} v_{\bar{i}} = \sum_{\bar{i}=1}^n b_{\bar{i}} v_{\bar{i}}$$

$$\Rightarrow 0 = \sum_{\bar{i}=1}^n a_{\bar{i}} v_{\bar{i}} - \sum_{\bar{i}=1}^n b_{\bar{i}} v_{\bar{i}} = \sum_{\bar{i}=1}^n (a_{\bar{i}} - b_{\bar{i}}) v_{\bar{i}}. \text{ B/c } v_1, \dots, v_n \text{ l.i.,}$$

$$0 = \sum_{\bar{i}=1}^n (a_{\bar{i}} - b_{\bar{i}}) v_{\bar{i}} \Rightarrow a_{\bar{i}} - b_{\bar{i}} = 0 \quad \forall \bar{i} \Rightarrow a_{\bar{i}} = b_{\bar{i}} \quad \forall \bar{i}.$$

Thus, w has a unique l.c. $a_1 v_1 + \dots + a_n v_n$. $\square$

$(\Leftarrow)$: Since every $w \in V$ is a l.c. of $v_1, \dots, v_n$,

$w \in \text{span}(v_1, \dots, v_n) \quad \forall w \in V$. Thus, $v_1, \dots, v_n$ spans V .

Also, $0 \in V$ has a l.c. $0 = 0 \cdot v_1 + \dots + 0 \cdot v_n$, which

is unique by assumption $\Rightarrow v_1, \dots, v_n$ l.i. by def.

Hence, $v_1, \dots, v_n$ l.i. & spans $V \Rightarrow v_1, \dots, v_n$ is a basis. $\square$

Rmk: In Unit 2, if $\beta = (v_1, \dots, v_n)$ is a basis of V , then coeffs $a_1, \dots, a_n$ of $w = a_1 v_1 + \dots + a_n v_n$ are the coords of w relative^{rel} to β : $[w]_\beta := \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n$.
(so relates $w \in V$ to $[w]_\beta \in \mathbb{R}^n$)

In many cases, we are interested in subsp $\text{span}(S)$ of $\mathbb{R}^n$, $\mathbb{R}[x]$, $M_{m \times n}(\mathbb{R})$ where S fin.

Q: Does $\text{span}(S)$ have a basis?

$\Downarrow$ Yes!

Prop 1.49: $V = \text{span}(S)$ (so V v.s., $S \subseteq V$) w/ S fin.

Then $\exists \beta \subseteq S$ basis for V (so V has a fin basis).

Pf: If S l.i., then S is a basis by def. So assume S l.d. Then $S \neq \emptyset$ by def, and S fin

$\Rightarrow$ def $\#S :=$ [#]number of elts of S , which is $< \infty$.

Assume by $\times$ that $\nexists$ basis $\beta \subseteq S$ for V . Then

$\forall T \subseteq S$ s.t. $\text{span}(T) = \text{span}(S) = V$, T is l.d. $(*)$

(or T is a basis for V). $S_1 := S$ l.d. so Prop 1.43

$\Rightarrow \exists v_1 \in S$ s.t. $S_2 := S_1 \setminus \{v_1\}$ spans V & l.d. by $(*)$

Similarly, $\exists v_2 \in S_2$ s.t. $S_3 := S_2 \setminus \{v_2\}$ l.d. & spans V .

Continuing this pattern (i.e., at i^{th} step, S_i l.d. and

spans V , so $\exists v_i \in S_i$ s.t. $S_{i+1} := S_i \setminus \{v_i\}$ l.d. and spans V , we get $S = S_1 = \{v_1, v_2, \dots\}$. But S fin, so $S = \{v_1, \dots, v_m\}$ where $m := \#S$. Then by ^{constr} construction $S_i = \{v_i, v_{i+1}, \dots, v_m\}$ l.d. $\forall 1 \leq i \leq m$, and $S_{m+1} = S_m \setminus \{v_m\} = \{v_m\} \setminus \{v_m\} = \emptyset$ l.i., ~~*~~ to S_j l.d. $\forall j \in \mathbb{Z}_{>0}$. Hence, $\exists \beta \subseteq S$ basis for V . $\square$

Rmk 1.50: Pf of Prop 1.49 says that if $V = \text{span}(S)$, $n = \#S < \infty$, & S l.d., then $\exists$ subsets $S = S_1 = \{v_1, \dots, v_n\} \supsetneq S_2 = \{v_2, \dots, v_n\} \supsetneq \dots \supsetneq S_\ell = \{v_\ell, \dots, v_n\}$ where $\ell \leq n+1$ (if $\ell = n+1$, then $S_\ell = \emptyset$), $V = \text{span}(S_i)$ $\forall i \leq \ell$, S_j l.d. $\forall j < \ell$, and S_ℓ l.i. Thus, S_ℓ is a basis for V .

Ex 1.51: Using Rmk 1.50, given

$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$, let's find a basis $\beta \subseteq S$ of $\text{span}(S) \subseteq \mathbb{R}^3$ then compute $\text{span}(S)$.

Notice that $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \stackrel{(*)}{=} 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$\Rightarrow$ let $T := S \setminus \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}$ then

$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \in \text{span}(T)$ by $(*) \xRightarrow{\text{Prop 1.42}} \text{span}(T) \stackrel{(\star)}{=} \text{span}(S)$.

Note that std basis $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{R}^3$ is l.i.

$\Rightarrow \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ l.i. by Cor 1.40. Since

$\text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \left\{ \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\} \not\ni \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, Lem 1.44

$\Rightarrow T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \cup \left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}$ l.i. Thus, $\beta := T$ is
 $\text{span}(\beta) = \text{span}(S)$ by $(\star)$

a basis of $\text{span}(S) \subseteq \mathbb{R}^3 = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$.

B/c $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \text{span}(\beta)$,

$\mathbb{R}^3 = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) \subseteq \text{span}(\beta) \subseteq \mathbb{R}^3$

$\Rightarrow \mathbb{R}^3 = \text{span}(\beta) \Rightarrow \beta$ basis of $\mathbb{R}^3$.

Rmk: In fact, every v.sp has a basis! Hard case
 is when $\nexists S$ fm w/ $V = \text{span}(S)$, as Pf relies on

Optional §1.7* of FIS. (Math Major Only!)

Rmk 1.52: In Ex 1.51, both β & std basis of $\mathbb{R}^3$ have 3 elts each. Is this a common pattern $\forall$ v.sp V ?

↓↓ Yes by two Results!

Thm 1.53 (Replacement Thm):

$V = \text{span}(G)$, $\#G = n < \infty$, $B \subseteq V$ l.i. w/

$\#B = m < \infty$. Then $m \leq n$ and $\exists H \subseteq G$ s.t.

$\#H = n - m$ and $B \cup H$ spans V .

Pf: Math-Major Only (in §1.6 of FIS) $\square$

Exercise 1.54: Use Replacement Thm to prove that if $V = \text{span}(G)$ w/ $\#G < \infty$, then $\forall B \subseteq V$ l.i. (w/o $\#B < \infty$ assumption), then $\#B < \infty$.

Cor 1.55: $V = \text{span}(\beta)$ w/ β fin basis. Then every basis $\tilde{\beta}$ for V is fin, and $\#\tilde{\beta} = \#\beta$.

Pf: $\tilde{\beta}$ is a basis of $V = \text{span}(\beta)$. Then $\tilde{\beta}$ l.i.

$\Rightarrow \underline{\#\tilde{\beta} \leq \#\beta}$ by Replacement Thm. Switch roles of β & $\tilde{\beta} \Rightarrow \underline{\#\beta \leq \#\tilde{\beta}} \Rightarrow \#\beta = \#\tilde{\beta}$. $\square$